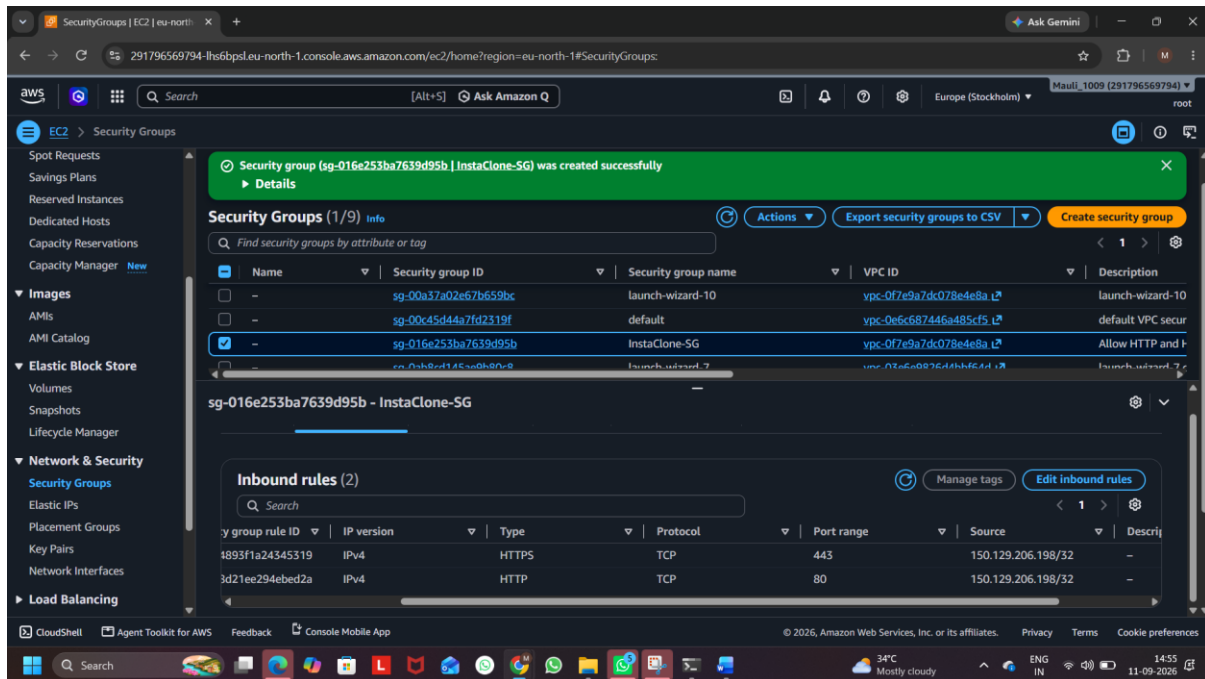


Cloud-Networking-Assignment-4

Q1.In AWS, create a new security group named InstaClone-SG and configure it to allow incoming HTTP (port 80) and HTTPS (port 443) traffic only from your own IP address.

Ans:



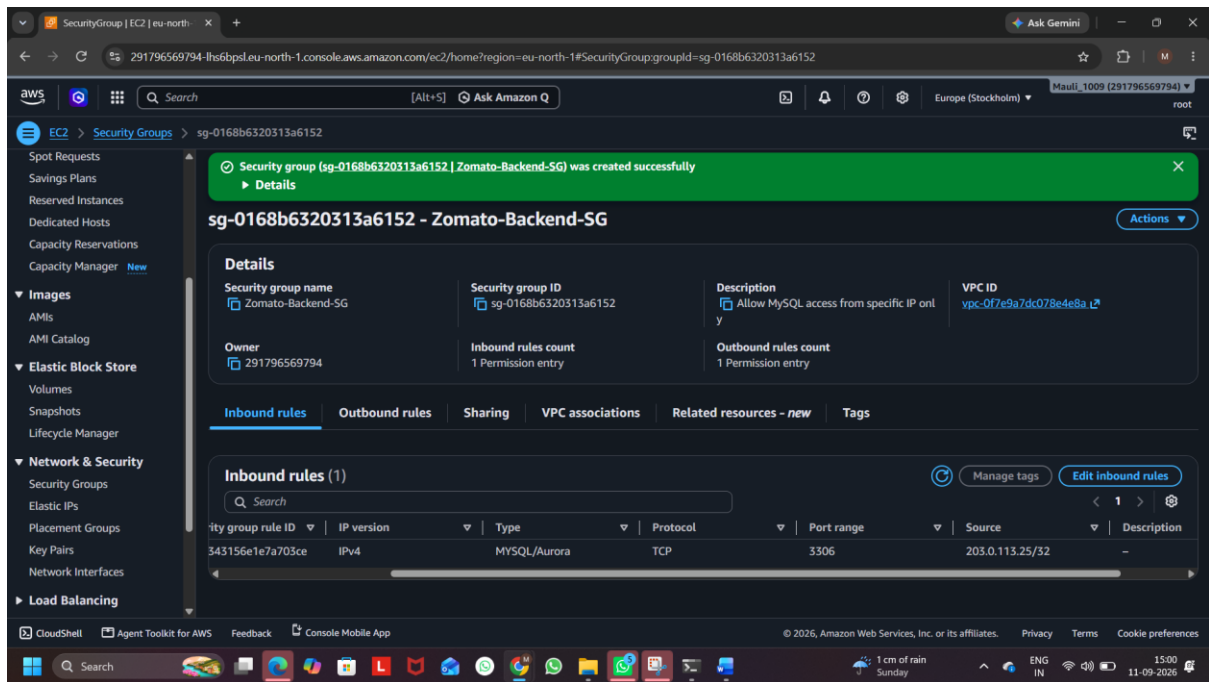
Screenshot: InstaClone-SG Inbound Rules

Created a new AWS Security Group named **InstaClone-SG** and configured inbound rules to allow **HTTP (port 80)** and **HTTPS (port 443)** traffic only from **My IP address**.

Q2.Set up a security group for a mock Zomato-style backend server that allows incoming MySQL (port 3306) connections only from a specific IP address and blocks all other inbound traffic. Hint: Use the 'Custom TCP' rule and specify the allowed IP in CIDR format.

Ans:

Created a security group named **Zomato-Backend-SG** and added a **Custom TCP** rule for **MySQL port 3306**. The rule allows connections only from the specified IP address in **CIDR format (/32)**.

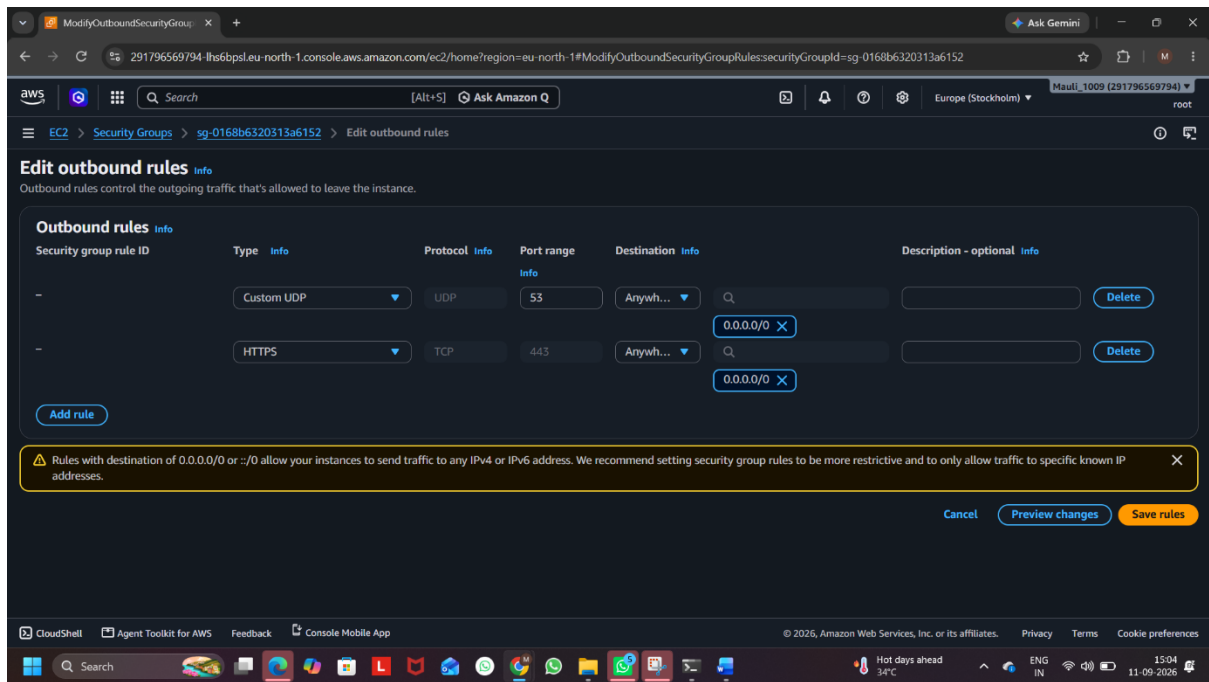


Screenshot: Zomato-Backend-SG Inbound Rules

Q3. Modify an existing security group to block all outbound traffic except for DNS (port 53) and HTTPS (port 443) to simulate a strict Paytm wallet server environment.

Ans:

Modified the existing security group by removing the default **All Traffic** outbound rule. Added outbound rules for **DNS (port 53)** and **HTTPS (port 443)** only.



Screenshot: Paytm Security Group Outbound Rules

Q4. Compare security groups and network ACLs by listing two key differences in how they control traffic for a Flipkart-like cloud application. Hint: Think about the level (instance vs subnet) and direction (inbound/outbound) each operates at.

Ans:

Security Groups	Network ACLs
1. Security Groups work at the instance level . They control traffic going to and from a specific EC2 instance.	1. Network ACLs work at the subnet level . They control traffic for all resources inside a subnet.
2. Security Groups are stateful and control inbound and outbound traffic. If an inbound request is allowed, the response is automatically allowed.	2. Network ACLs are stateless and have separate rules for inbound and outbound traffic.

Example for Flipkart:

For a Flipkart-like application, a **Security Group** can protect a particular web server, while a **Network ACL** can provide an additional traffic-control layer for the entire subnet.